





The best walking shoes are suitable for me
day.

Rowemercer

Sound Amplitude Modulated Laser Beam.

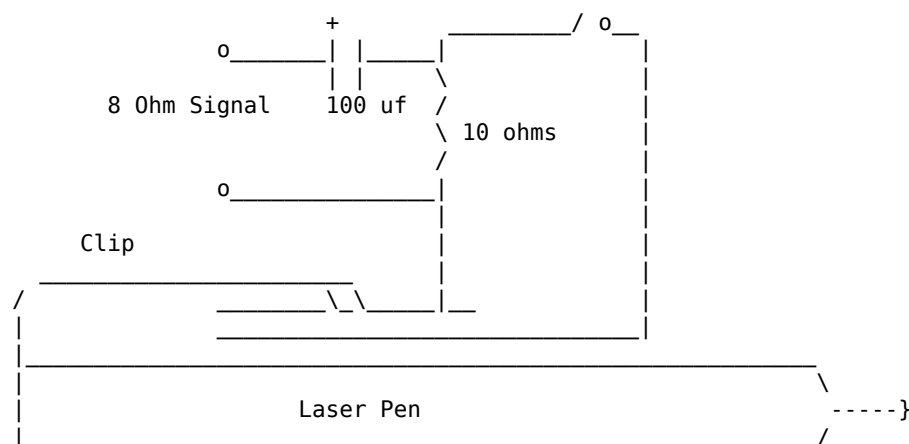
Most Laser Pens can easily be adapted to send a sound amplitude modulated beam. The ones that use the pocket clip as a switch are the easiest to use for this purpose, but the ones with a button can still be used but not quite as conveniently.

With the modulated Laser Beam, a signal can easily be sent many many times farther than what can easily be done with the modulated LED.

The trick to modulating a laser is to put a 10 ohm resistor in series with the batteries and couple a sound signal across the resistor through a 100 uf capacitor from an 8 ohm audio source. The source can be the earphone output of a small amplifier driven by a microphone or the earphone output of a small transistor radio.

The kind of laser pen that uses the pocket clip as the switch is the easiest to use since the circuit can be connected to a small piece of two sided copper clad PC board material. The PC board material can then be easily slipped under the clip. This puts the circuit in series with the laser power supply (batteries).

The kind of laser pen that has a button in the side can also be used but you will need to remove the back end of the laser pen and connect the circuit between the center battery terminal and the laser body. A bit of tape and ingenuity etc. will be required. The button on the side of the laser can then be held closed with a piece of tape or rubber band around the laser.



Warning: The signal should be very low level as it is may be possible to blow the laser with too much signal input. The right amount of signal will sound very soft when driving just the radio

speaker but will be enough to modulate the laser well. The light from the laser should flicker only slightly when modulated.

The receiver can simply be a photocell plugged into a small audio amplifier. Both are available at Radio Shack. They have a tiny palm sized audio amplifier. This type of amp could also be used to drive the laser transmitter from a microphone.

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